

MSCI151: Tools and Techniques for Business Analytics

Coursework 2: Lent Term 2022/23

Instructions

- This coursework is worth 10% of the credit for this module.
- All questions should be answered.
- Your work must be submitted via Moodle by **Thursday February 16, 2023 at 14:00**.
The link for submission is <https://modules.lancaster.ac.uk/mod/assign/view.php?id=2035628>. Late submissions will be subject to standard departmental penalties for late-submitted work.
- Your answers can be either handwritten or typed using Microsoft Word. If your answers are handwritten, then they should be scanned and submitted via Moodle (using the link above) and you must make sure the scanned files are easily readable. If your answers are typed, you must make sure that any mathematical notation is well-presented and easily understandable, and any diagrams should be inserted as pictures.
- If you make mistakes in one part of a question, you may still receive credit for correct methods/calculations in later parts even if your answers are affected by the earlier mistakes (i.e. you may receive 'error carried forward' marks).
- If you have any questions about this coursework, please post them on the 'Questions and Answers' forum: <https://modules.lancaster.ac.uk/course/view.php?id=37604>. Note that lecturers may not have enough time to respond to questions that are asked near the submission deadline – so don't leave your questions too late.
- This is an **individual** coursework assignment, so you should avoid collaborating with others. Collusion between students and plagiarism are regarded as serious offences. Information about the university's plagiarism framework can be found here: <https://modules.lancaster.ac.uk/mod/resource/view.php?id=1384800>.

Question 1

Consider a simple simulation of the weather in which the weather conditions on the next day depend probabilistically on the weather conditions on the present day, as shown by the table below:

		Weather conditions on next day		
		Sunny	Overcast	Rainy
Weather conditions on present day	Sunny	0.4	0.35	0.25
	Overcast	0.15	0.5	0.35
	Rainy	0.25	0.45	0.3

The table above should be interpreted as follows: if the weather on the present day is sunny, then the weather on the next day can be either sunny (with probability 0.4), overcast (with probability 0.35) or rainy (with probability 0.25). Similar interpretations also apply to the other two rows in the table.

- (a) Suppose the weather on September 1st is sunny. Use **dynamic simulation** to simulate the weather conditions on the next 10 days, from September 2nd to September 11th (inclusive). You should carry out the simulation using the random number table provided on Moodle (see <https://modules.lancaster.ac.uk/mod/resource/view.php?id=2047288>) and make it clear which ranges of random numbers you are using for the different possible outcomes (e.g. if the weather on the present day is sunny, the range 01-40 could be used for sunny weather on the next day, etc). You should also make it clear which row(s) or column(s) of the random number table you are using for each of the distributions that you sample from. **Note that you may lose marks if you do not make it clear how you are sampling digits from the random number table (i.e. which rows/columns you are using), and also how you are converting random digits into outcomes.**

(20 marks)

- (b) Use the results of your simulation in part (a) to estimate the number of days during the whole year (365 days) on which the weather conditions are unchanged from the previous day.

(5 marks)

- (c) Do you think your estimate in part (b) could have been affected by start-up bias? Explain your answer as fully as possible.

(5 marks)

Total marks for Question 1: 30

Question 2

At a local self-service petrol station, customers arrive at an average rate of 0.25 per minute. You may assume that the inter-arrival times are exponentially distributed. There is only a single pump available to serve all customers, so that the system operates as a single-server queue. The time required for customers to pull up, fill up with petrol, pay and leave can be modelled using a continuous uniform distribution with a minimum of 2 minutes and a maximum of 6 minutes. If a new customer arrives to find that two customers are already at the station (with one using the pump and the other waiting in the queue), then the new customer leaves immediately to find an alternative service and we say that this customer is 'lost' to the station.

- (a) Use **discrete event simulation** to simulate the period from 5:00pm to 5:30pm. You should begin your simulation by assuming there are no customers at the station at 5:00pm. Also, in this question (unlike the previous one) you are given some specific information about how to carry out the simulation using the random number table:
- To sample the inter-arrival times, use the formula $(-1/\lambda) \times \ln(R)$, where λ is the arrival rate and R is uniformly distributed between 0 and 1. You should sample the values of R using **Row 5** of the random number table provided on Moodle (65 24 30 11 etc.) and read

from left to right, using 4 digits for each required value of R . You may continue and use Row 6 of the table if you need to.

- To sample the service times, use the formula $a + (b - a) \times R$ where a and b are the minimum and maximum values respectively and R is uniformly distributed between 0 and 1. You should sample the values of R using **Row 7** of the random number table provided on Moodle (74 93 50 70 etc.) and read from left to right, using 4 digits for each required value of R . You may continue and use Row 8 of the table if you need to.

You should round all arrival times and service times to **2 decimal places**.

To help you, the first parts of the state list and event list are completed for you – see below. (In these results, 5:00pm is counted as time zero.)

State List			<i>Statistics</i>		Event List			
Time (mins)	In queue	Using pump	Waiting time	Lost? (Y/N)	R.N.	Event	Time (mins)	Done ?
0.00	-	-	-	-	6524	Arrival C1	1.71	✓
1.71	-	C1	0.00	No				

Note that you will lose marks if you do not follow the specific instructions for sampling inter-arrival times and service times given above.

(20 marks)

(b) Use the results of your simulation to estimate:

- the average amount of time that customers have to wait before using the pump (ignoring any customers that are lost to the station),
- the expected number of customers lost per hour.

(10 marks)

Total marks for Question 2: 30

Question 3

Machines in a factory break down at an average rate of 3 per day. When a machine breaks down, it enters a queue to be repaired. There is a single repairperson on duty, who repairs machines at an average rate of 5 per day. Assume that the process of machines breaking down and being repaired can be modelled as an M/M/1 queue with an arrival rate of $\lambda = 3$ per day and a service rate of $\mu = 5$ per day. Although the factory is only open for 8 hours per day, the state of the system

at the beginning of any particular day is the same as the state at the end of the previous day, so that the system effectively evolves in a continuous, uninterrupted way over time.

- (a) Calculate the probability that **4 or more** machines break down during a single day.
(4 marks)
- (b) Given that a working day lasts 8 hours, calculate the average amount of time per day that the repairperson spends working (i.e. repairing machines).
(4 marks)
- (c) Calculate (i) the expected number of machines waiting in the queue to be repaired at an arbitrary point in time (not including the machine currently repaired, if there is one); (ii) the expected amount of time that a machine must wait in the queue before the repairperson begins work on it.
(8 marks)
- (d) The factory owners pay the repairperson £120 per day. During the time that a machine is broken (including the time spent in repair), the factory incurs costs at a rate of £95 per day due to loss of productivity. The owners are considering hiring an extra repairperson, who would be paid at the same rate as the current one (you may also assume that they can repair machines at the same rate). Do you think it would be profitable for them to hire the extra repairperson? Show your calculations and also state any additional assumptions that you make.
(10 marks)
- (e) After some further analysis, the factory owners believe that the amount of time that the repairperson spends repairing a machine actually has a triangular distribution with minimum value $a = 0.8$ hours, maximum value $b = 2.7$ hours and mode $c = 1.3$ hours. Given this new information, they believe that the system should be modelled as an M/G/1 queue rather than M/M/1. (Assume that they continue to use a single repairperson.) Recalculate the performance measures in part (c) for the new M/G/1 system. [Hint: look up the properties of the triangular distribution online.]
(10 marks)
- (f) State **two** standard assumptions of an M/G/1 queue and comment on whether these are likely to be realistic for the system considered in this question.
(4 marks)

Total marks for Question 3: 40